Naveed Ahmed

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Professional Summary

Full-Stack MERN Developer with 3+ years of professional experience building scalable web applications, enterprise software, and production-grade AI-powered solutions. Skilled in React.js, Node.js, Express.js, MongoDB, MySQL, AWS, Docker, and Redis, with hands-on experience building AI/ML-powered systems, LLM integration, Retrieval-Augmented Generation (RAG), AI agents, tool calling, structured LLM outputs, and real-time AI streaming. Proven experience leading development teams, driving technical decisions, and delivering high-quality products from concept to production.

Technical Skills

Languages: HTML, CSS, JavaScript, TypeScript, Python
Frameworks & Libraries: React.js, Node.js, Express.js, Redux Toolkit, Bootstrap, Tailwind CSS, NestJS, Next.js, Django
AI Engineering: LLM Integration, Structured Output, RAG, Embeddings, SSE Streaming, Tool Calling, AI Agents, Guardrails, Recommendations
Databases: MongoDB, MySQL, PostgreSQL, Redis
Message Broker & Caching: RabbitMQ, Redis
Tools & Technologies: Git, GitHub, Docker, AWS, Postman, JIRA, ClickUp, CI/CD Pipelines, Claude, Cursor

Experience

TheCloudOps — September 2025 – Present

Senior MERN Stack Developer / Team Lead — *Lahore, Punjab*

- Leading the end-to-end development of **Private Jet Aviation Charter Platform**, a full-stack aviation management platform built with the MERN stack, supporting a real-time flight booking and management system.
- Driving full-stack development of a **Music Platform**, an ad-supported music lyrics platform with an 11M+ document catalog, owning AI-powered ad targeting, header-bidding integration, SEO optimization, and large-scale MongoDB migrations.
- Implemented and integrated AI-powered features, including AI-driven advertising and intelligent application capabilities, leveraging modern AI/LLM technologies such as RAG, embeddings, vector search, tool/function calling, and AI agents.
- Leading and mentoring development team members by conducting code reviews, providing technical guidance, breaking down complex requirements into actionable tasks, and ensuring development follows scalable and maintainable engineering practices.
- Acting as the primary technical point of contact for clients, gathering requirements, translating business needs into technical solutions, and ensuring timely delivery.

OIOI Group — October 2024 – August 2025

MERN Stack Developer — *Lahore, Punjab*

- Developed and maintained a Bike Inventory Management System designed to streamline warehouse operations, inventory tracking, stock movement, and transfer management.
- Designed and implemented secure CRUD APIs for inventory management, ensuring reliable creation, updating, retrieval, and deletion of inventory records with appropriate validation and access controls.
- Collaborated with cross-functional teams to understand business requirements, troubleshoot technical issues, and deliver maintainable, production-ready software solutions.
- Deployed and managed cloud-based solutions on AWS to ensure scalability and high availability.

KOT Enterprises — November 2023 – September 2024

Junior MERN Stack Developer — *Lahore, Punjab*

- Developed and maintained backend services and RESTful APIs using Node.js and Express.js, supporting scalable and maintainable web application architectures.
- Designed and implemented MongoDB database schemas, models, and queries, ensuring efficient data storage, retrieval, and management across application modules.
- Built and integrated RESTful APIs for core application features, implementing request validation, error handling, authentication, and reliable communication between frontend and backend systems.
- Contributed to the development of responsive and interactive frontend applications using React.js, JavaScript, and Tailwind CSS.

Featured Projects

Private Jet Aviation Platform | *React, TypeScript, Node.js, Express.js, MySQL, Stripe, AWS* **Sep 2025 – Present**

- Developed a full-stack private jet charter marketplace connecting operators, brokers, and flyers.
- Engineered Stripe payments end-to-end: authorize-then-capture checkout with SHA-256 idempotency keys, Connect Express operator payouts with instant-payout auto-fallback, tokenized broker invoicing.
- Designed a concurrency-safe booking engine with 15-minute soft locks preventing double-booking, a 6-step checkout (in-app PDF contract signing, card/wire/balance payments), and owner-approval payment holds.

Ad-Supported Music Platform | *Next.js, TypeScript, NestJS, MongoDB, Redis, BullMQ* **December 2025 – Present**

- Developed an ad-supported music lyrics platform serving an 11M+ document catalog (songs, artists, albums) with SEO-optimized ISR/RSC pages, JSON-LD structured data, and Core Web Vitals tuning.
- Engineered AI content pipelines (song meanings, news) using Gemini, Vertex AI batch jobs, and BullMQ; implemented Google Ad Manager key-value targeting and rolled out PubGuru header bidding to production in phases.
- Built an admin CMS with role-based access control, AI article generation and scheduling, a batch-job operations console, and a UGC moderation suite.
- Cut API query times by 90% (300ms to 50ms) by consolidating a 7M+ document catalog from scattered staging collections into 3 core MongoDB collections with indexed ID-based lookups.

AI-Powered Knowledge & Automation Platform | *React, Node, TypeScript, Gemini, RAG* **May 2026 – Present**

- Developed and contributed to a production-oriented AI platform, progressively implementing LLM capabilities including intent detection, structured outputs, RAG, streaming, tool calling, and AI agents.
- Designed and implemented an end-to-end Retrieval-Augmented Generation (RAG) pipeline, including document chunking, embeddings, semantic retrieval, relevance thresholds, grounded response generation, and citations.
- Improved RAG retrieval quality through heading-aware chunking, query rewriting, retrieval memory, and lazy query rewriting, ensuring responses are based on relevant knowledge rather than blindly retrieved content.
- Developed a bounded multi-step AI agent architecture with execution limits, no-progress detection, and step-level tracing for reliable and observable agent execution.
- Implemented AI guardrails and cost-control mechanisms, including token budgets, usage monitoring, backend-secured API credentials, and protections against uncontrolled model execution.

Education

Government College University (GCU), Lahore **2016 – 2020**

BSc. (Hons.) Physics *Lahore, Pakistan*

- Authored a B.Sc. (Hons.) thesis, *"Different Attempts To Make Quantum Computers and Computation a Reality,"* analyzing how computers process and represent information—from classical architecture (binary, logic gates, transistors, CPU, memory, data encoding) to emerging quantum models (qubits, superposition, entanglement, quantum logic gates).